

Vol 5 Architectural Scaffold v0.1 — Quantum Mechanics from D_IV⁵ (Pedagogical Bridge)

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Filed: 2026-05-23 Saturday (Lyra, Wave 2 scaffold) Companion to: INDEX.md (chapter outline + reader target)

Anchoring BST results (existing coverage by chapter)

Ch 1 From Substrate to Standard Hilbert Space (~30%)

Anchors: Vol 1 Ch 2 + T2428 + T2429 - $L^2(\mathbb{R}^3)$ standard Hilbert space emerges as macroscopic limit of Bergman $H^2(D_{IV}^5)$ - Substrate-tick $GF(128)^k$ discretization (T2429 RS Reed-Solomon) - L^2 -section equivariant complement (T2430) - Gap: explicit emergence-limit derivation (multi-week pedagogical work)

Ch 2 Position + Momentum + Heisenberg (~50%)

Anchors: T2419 + T2422 + Vol 0 Ch 4 §4.2 v0.5 coset Cartan - T2419 position M_z as multiplication operator on Bergman H^2 - T2422 momentum P_z as Wirtinger derivative - Coset translation directions $m \subset \mathfrak{so}(5,2)/(\mathfrak{so}(5) \oplus \mathfrak{so}(2))$ substrate-anchor (Vol 0 Ch 4 §4.2 v0.5) - $[M_z, P_z] = -i$ canonical commutator via Bergman kernel reproducing property

Ch 3 Angular Momentum + Spin (~50%)

Anchors: T2425 + T2421 + $\text{Pin}(2)$ Z_2 - T2425 angular momentum $L = 10 \text{ SO}(5)$ rotation generators - T2421 spin S via $\text{SO}(5) \times \text{SO}(2)$ K-type representation - $\text{Pin}(2)$ Z_2 grading per T1925 rank=2 (Paper #133 spin-statistics) - Half-integer spin = $\text{SO}(2)$ spinor half-weight (T2471 chirality = $\text{Pin}(2)$ Z_2)

Ch 4 Schrödinger Equation (~40%)

Anchors: Vol 1 Ch 7 + T2438 + Elie K52a S29 - $i\hbar \partial|\psi\rangle/\partial t = H_{\text{sub}} |\psi\rangle$ with $H_{\text{sub}} = \text{Casimir on } L^2(D_{IV}^5; L_\lambda)$ - Standard time-dependent + time-independent Schrödinger from substrate - T2473 energy conservation via $\text{SO}_0(5,2)$ time-translation invariance

Ch 5 Heisenberg Picture + Path Integral (~40%)

Anchors: Vol 1 Ch 7 + Ch 9 + T2457 - Heisenberg $dO/dt = (i/\hbar)[H_{\text{sub}}, O]$ for operator zoo - Path integral via substrate-tick $GF(128)^k$ finite-step sum (Vol 1 Ch

7 + Ch 9) - T2457 Bergman structural-role-of Feynman propagator - Gap: explicit Feynman path integral derivation from substrate ticks (multi-week)

Ch 6 Hydrogen Atom + Atomic Spectra (~60%)

Anchors: Vol 1 Ch 5 Casimir + Toy 541 + Vol 3 Ch 7 atomic - Standard hydrogen Rydberg formula $E_n = -13.6 \text{ eV}/n^2$ - BST: hydrogen energy levels from substrate-natural Casimir spectrum - Selection rules from substrate K-type transition allowed/forbidden

Ch 7 Born Rule + Measurement (~70%)

Anchors: K67 RATIFIED + T2479 + T2469 - K67 Born = Bergman reproducing-kernel evaluation (RATIFIED Tuesday) - T2479 POVM substrate-derivation (Saturday) - T2469 SCMP operational theorem (Friday) - Standard projective measurement + POVMs as substrate-cartography readings

Ch 8 Bell-CHSH + Quantum Correlations (~80%)

Anchors: T2399 + Calibration #17 + T2469 SCMP - Standard Bell-CHSH inequality + Tsirelson bound - BST T2399 trace-level 126/16 substrate prediction - Sub-Tsirelson deviation $1/8 = 1/2^{N_c}$ falsifier (T2469 SCMP) - Vienna/Caltech/Munich Bell experiment design \$300-500K

Ch 9 Identical Particles + Spin-Statistics (~70%)

Anchors: Paper #133 v0.2 + T2471 + Pin(2) Z_2 - Pauli exclusion + boson/fermion partition from Pin(2) Z_2 grading - $\gamma^5 = \text{Pin}(2) Z_2$ grading object (T2471 Friday) - Spin-statistics without relativistic invariance assumption - Cross-link to Vol 1 Ch 4 discrete symmetries

Ch 10 Decoherence + Classical Limit (~40%)

Anchors: T2469 SCMP + Vol 4 Ch 3 boundary - T2469 SCMP framework: classical limit = substrate-ensemble marginalization - Substrate-coupled observer bandwidth bounded $\rightarrow$ environment-induced superselection - Cross-link to Zurek decoherence + Wojciech-Halliwel classical-emergence - Gap: full quantitative decoherence-rate derivation (multi-month)

Ch 11 POVMs + Quantum Information Basics (~50%)

Anchors: T2479 + K67 extension + SP-31 #283 - T2479 POVM substrate-derivation closure (Saturday) - Naimark dilation + Kraus operators as substrate-cartography - Quantum information primitives (no-cloning + entanglement + teleportation) cross-link to Vol 7

Ch 12 Pedagogical Bridge Synthesis (~30%)

Anchors: Vol 1 + Vol 0 cross-references - Standard QM curriculum recovery (Griffiths / Sakurai / Shankar) - Cross-curriculum reading guide: where each standard QM concept lives in BST volumes - Multi-volume index for QM-track readers

~50% existing BST coverage across Vol 5 chapters; 50% pedagogical-bridge prose + worked-example development pending.

Cross-volume reference map

Inputs from Vol 0: D_{IV}^5 + five integers + operator zoo (12/14 STRUCTURALLY VERIFIED) Inputs from Vol 1: Bergman Hilbert space + Casimir + Dynamics + Operator zoo (6/6) Outputs to: Vol 7 (Bell + POVMs + Information Theory), Vol 8 (substrate-coupled observers), Vol 9 (many-body QM)

— Vol 5 Architectural Scaffold v0.1, Saturday 2026-05-23 Wave 2 scaffold